Course 2022 | Semester I / II | Theory | 4 Credits | 60 Contact Hours

Engineering Mechanics and Design Thinking

Analyse force systems, apply equilibrium conditions, determine centroid & moment of inertia, and embrace design thinking for engineering innovation.

PROGRAMME Mechanical Engineering	COURSE CODE 2022
SEMESTER I / II	COURSE TYPE Theory
L:T:P:S 4:0:0:0	CREDITS 4
CIE MARKS 40	ESE MARKS 60

📄 Official Source Declaration

This handbook is based on the official **Kerala Polytechnic Revision 2026** syllabus for **Course 2022 — Engineering Mechanics and Design Thinking**. The syllabus document (file: `2022.pdf`) has been used as the sole authoritative source for all module content, topics, subtopics, learning outcomes, teaching scheme, assessment pattern, and references.

> ✅ **Verified consistency:** All four modules, instructional hours, CO–PO mapping, CIE/ESE scheme, and self-learning activities have been extracted and presented as per the official document. No external internet research was used.

> 📌 **Note on source integrity:** The PDF contains minor inconsistencies in the presentation of self-learning activity hours (listed twice with slightly different phrasing) and the assessment table formatting. This handbook preserves the official content and presents it in a clear, student-friendly layout. The total self-learning hours (87.5) are taken from the official table.

📊 Course Dashboard

60
Contact Hours

4
Credits

4:0:0:0
L:T:P:S

40
CIE Marks

60
ESE Marks

87.5
Self-Learning Hours

Course Objectives

1. To impart the knowledge and skills that enable the student to predict the effects of force and motion.
2. To appreciate the concepts of design thinking and its salient applications in the engineering field.

Course Pre-requisites

Pre-requisite Courses

Topic	Course Code	Course Title	Semester
Trigonometry	1002	Fundamentals of Engineering Mathematics	I
Scalar, Vector quantities, triangular and parallelogram law of forces	2002A	Engineering Physics for mechanical, structural and industrial applications	I/II
Integration	2001A	Mathematics for Industrial Engineering	II

Teaching & Assessment Scheme

Teaching and Assessment Scheme

Teaching Scheme/Week	Self-Learning (SS)/Week	Total Hours/Semester	Credits	CIE (Theory)	ESE (Theory)	Total
3	5.5	60	4	40	60	100

L: Lecture (1 hr = 1 credit) · **T:** Tutorial (1 hr = 1 credit) · **P:** Practical (1 hr = 0.5 credit) · **S:** Project (1 hr = 0.5 credit) · **SS:** Self-Learning · **CIE:** Continuous Internal Evaluation · **ESE:** End Semester Examination

🧭 How to Use This Handbook

1
📖 **Understand**
Read each module & topic with deep explanations.

2
📐 **Visualise**
Study labelled diagrams & formula cards.

3
✏️ **Apply**
Solve worked examples & use interactive calculators.

4
✅ **Revise**
Practice Q&A, model paper, and quick-revision cards.

🎯 Course Outcomes

Course Outcomes (CO) with Bloom's Taxonomy Level

CO	Course Outcome	Bloom's Level
CO1	Identify the force systems for given conditions by applying the basics of mechanics	Apply
CO2	Apply conditions of static equilibrium to determine unknown force(s) of different structural elements.	Apply
CO3	Solve problems involving rigid bodies by applying the properties of distributed areas and masses.	Apply
CO4	Appreciate the concept of design thinking and its salient applications in the engineering field.	Understand

CO–PO Mapping

Course Articulation Matrix

CO \ PO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11
CO1	3	3	—	—	2	—	—	—	—	—	—
CO2	3	3	2	—	2	—	—	—	—	—	—
CO3	3	3	2	2	2	—	—	—	—	—	—
CO4	—	—	—	—	—	2	2	—	2	2	2
Total	9	9	4	2	6	2	2	—	2	2	2

Legends: 1 — Low; 2 — Medium; 3 — High; No Correlation — "—"

📋 Curriculum Coverage Map

Module-wise coverage with instructional hours and CO mapping

Module	Title	Major Topics	Hours	CO
I	Force Systems	Define force systems, resolve forces, resultant, moment of a force, free body concept, Lami's theorem	11	CO1
II	Equilibrium & Friction	Beams, loads, supports, reactions (simply supported & cantilever), principles of friction	12	CO2
III	Distributed Areas & Masses	Centroid, centre of gravity, moment of inertia, composite figures & solids, polar moment of inertia	12	CO3
IV	Design Thinking	Concept, stages (Empathize, Define, Ideate, Prototype, Test), techniques, tools, applications	10	CO4

🧩 Module Roadmap

I
Force Systems
Resolution, resultant, moment, FBD, Lami

II
Equilibrium & Friction
Beam reactions, loads, supports, friction

III
Distributed Areas & Masses
Centroid, CG, moment of inertia

IV
Design Thinking
Stages, tools, prototyping, applications

MODULE I

11 hrs · CO1 · Apply

Force Systems

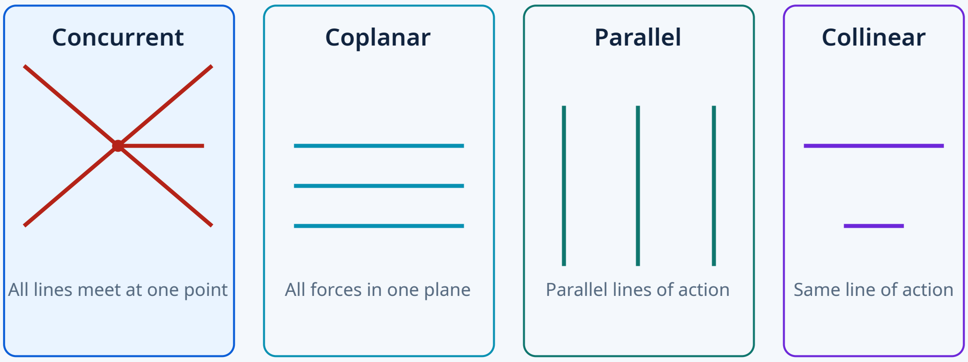
Define force systems · Resolution & resultant · Moment of force · Free body concept · Lami's theorem

This module builds the foundation of engineering mechanics: understanding how forces act, how to break them into components, find their combined effect, compute moments, and analyse equilibrium using free body diagrams and Lami's theorem.

▼Define different force systems

Force: A push or pull that changes or tends to change the state of rest or motion of a body. Force is a vector quantity — it has magnitude, direction, and point of application.
Force Systems: A set of forces acting on a body can be classified based on their line of action and point of concurrence.

- **Concurrent forces:** Lines of action meet at a single point.
- **Coplanar forces:** All forces lie in the same plane.
- **Collinear forces:** Forces act along the same line.
- **Parallel forces:** Lines of action are parallel (like or unlike).
- **Non-concurrent, non-parallel forces:** General force system.



Types of force systems: concurrent, coplanar, parallel, and collinear.

> **Malayalam support:** ബലം (Force) എന്നത് ഒരു വസ്തുവിന്റെ നിലയിലോ ചലനത്തിലോ മാറ്റം വരുത്തുന്ന പുഷ് അല്ലെങ്കിൽ പുൾ ആണ്. ബലങ്ങളെ അവയുടെ പ്രവർത്തനരേഖയുടെ അടിസ്ഥാനത്തിൽ concurrent, coplanar, parallel, collinear എന്നിങ്ങനെ തരംതിരിക്കുന്നു.

▾Resolve forces and find resultant of different force systems

Resolution of forces: Breaking a single force into two or more components, typically along orthogonal (perpendicular) axes. The most common method is resolving into horizontal and vertical components.

Resolution formulae:

$$F_x = F \cdot \cos\theta \quad \text{(horizontal component)}$$

$$F_y = F \cdot \sin\theta \quad \text{(vertical component)}$$

where θ is the angle with the horizontal axis.

Resultant force: A single force that has the same effect as the combined action of all forces in a system. For a system of coplanar forces:

$$R = \sqrt{(\Sigma F_x^2 + \Sigma F_y^2)}$$

$$\alpha = \tan^{-1}(\Sigma F_y / \Sigma F_x)$$

where R is the magnitude, α is the angle with the horizontal.

Worked example:
Given: $F_1 = 100$ N at 30°, $F_2 = 150$ N at 120°.
Required: Resultant magnitude and direction.
Calculation:

$\Sigma F_x = 100 \cos 30° + 150 \cos 120° = 86.60 - 75 = 11.60$ N

$\Sigma F_y = 100 \sin 30° + 150 \sin 120° = 50 + 129.90 = 179.90$ N

$R = \sqrt{(11.60^2 + 179.90^2)} = 180.27$ N

$\alpha = \tan^{-1}(179.90 / 11.60) = 86.3°$

Answer: R = 180.27 N at 86.3° from horizontal.

Malayalam support: ഒരു ബലത്തെ അതിന്റെ ഘടകങ്ങളാക്കി വിഭജിക്കുന്നതിനെ ബലവിഭജനം (resolution) എന്നു പറയുന്നു. എല്ലാ ബലങ്ങളുടെയും ആകെ ഫലമായി പ്രവർത്തിക്കുന്ന ഒറ്റ ബലത്തെ resultant എന്നു വിളിക്കുന്നു.

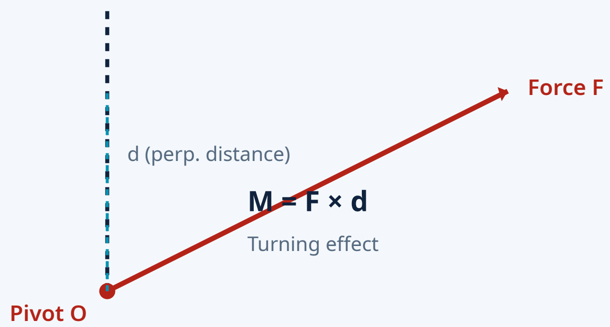
▼Compute moment of a force

Moment of a force (also called torque) is the turning effect of a force about a point or axis. It is the product of the force and the perpendicular distance from the point to the line of action of the force.

Moment (M) = F × d

F = force (N), d = perpendicular distance (m), M = moment (N·m)

Sign convention: Clockwise moments are often taken as negative and counter-clockwise as positive (or vice versa, depending on the convention).



Moment = Force × perpendicular distance from the pivot.

Worked example:
Given: A force of 50 N is applied at 2.5 m from a pivot.
Required: Moment about the pivot.
Calculation: M = 50 × 2.5 = 125 N·m
Answer: 125 N·m (counter-clockwise if direction is as shown).

Malayalam support: ഒരു ബലത്തിന്റെ മൊമന്റ് എന്നത് ആ ബലം ഒരു ബിന്ദുവിനെ ആപേക്ഷിച്ച് ഉണ്ടാക്കുന്ന തിരിക്കൽ ഫലമാണ്. ബലത്തിന്റെയും ലംബദൂരത്തിന്റെയും ഗുണനഫലമാണ് മൊമന്റ്.

▼Solve simple problems from free body concept and Lami's theorem

Free Body Diagram (FBD): A diagram showing a body isolated from its surroundings with all forces acting on it clearly marked. This is the first and most important step in solving any mechanics problem.

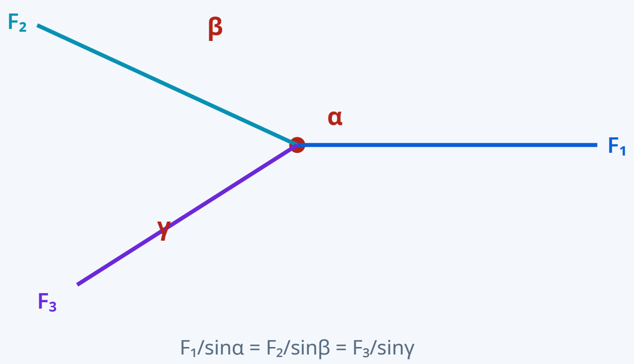
- Draw the body as a dot or simple shape.
- Show all external forces (gravity, applied loads, reactions, friction).
- Indicate known and unknown forces with labels.
- Choose a convenient coordinate system.

Lami's theorem: If three forces acting at a point are in equilibrium, then each force is proportional to the sine of the angle between the other two forces.

Lami's theorem:

$F_1 / \sin\alpha = F_2 / \sin\beta = F_3 / \sin\gamma$

where α, β, γ are the angles opposite to F_1, F_2, F_3 respectively.



Three forces in equilibrium — Lami's theorem relates each force to the sine of the opposite angle.

Worked example (Lami's theorem):
Given: Three forces: F_1 = 100 N, F_2 = 80 N, angles α = 120°, β = 100°, γ = 140°.
Required: Verify equilibrium using Lami's theorem.
Calculation:

$F_1 / \sin\alpha = 100 / \sin 120° = 100 / 0.866 = 115.47$

$F_2 / \sin\beta = 80 / \sin 100° = 80 / 0.985 = 81.22$

$F_3 / \sin\gamma = F_3 / \sin 140° = F_3 / 0.643$

For equilibrium, $115.47 = 81.22 = F_3/0.643 \Rightarrow F_3 = 74.23$ N

Answer: The system is in equilibrium when F_3 = 74.23 N.

Malayalam support: Free Body Diagram (FBD) ൽ വസ്തുവിനെ ഒറ്റപ്പെടുത്തി അതിൽ പ്രവർത്തിക്കുന്ന എല്ലാ ബലങ്ങളും അടയാളപ്പെടുത്തുന്നു. Lami's theorem മൂന്ന് ബലങ്ങൾ സന്തുലിതാവസ്ഥയിലായിരിക്കുമ്പോൾ ബലങ്ങളും അവയ്ക്കെതിരായ കോണുകളുടെ sine-ഉം തമ്മിലുള്ള ബന്ധം നൽകുന്നു.

💡 **Module I key tip:** Always start with a clear Free Body Diagram. Resolution and Lami's theorem are powerful tools — practice with varied angles and force magnitudes to build speed and accuracy.

MODULE II

12 hrs | CO2 | Apply

Equilibrium & Friction

Beams, loads & supports | Simply supported reactions | Cantilever reactions | Friction principles

This module applies static equilibrium conditions to determine unknown forces in structural elements. You will learn to identify beam types, calculate support reactions under various loads, and understand the principles of friction.

▼Identify different types of beams, loads and supports

Beam: A structural element that carries loads primarily by bending. Common types:

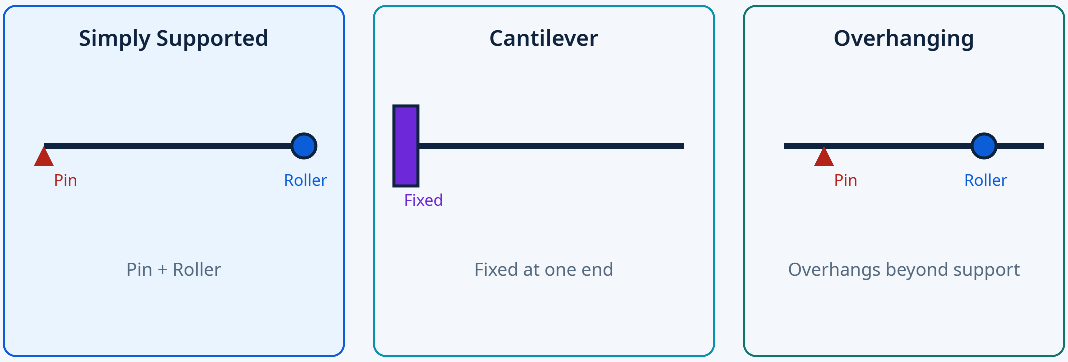
- **Simply supported beam:** Supported at both ends (one pin, one roller).
- **Cantilever beam:** Fixed at one end, free at the other.
- **Overhanging beam:** Extends beyond one or both supports.
- **Fixed beam:** Fixed at both ends.
- **Continuous beam:** More than two supports.

Loads:

- **Point load (concentrated):** Acts at a single point.
- **Uniformly Distributed Load (UDL):** Spread evenly over a length (kN/m).
- **Uniformly Varying Load (UVL):** Varies linearly along the length.

Supports:

- **Pin support:** Restrains translation in all directions; provides horizontal and vertical reactions.
- **Roller support:** Restrains translation perpendicular to the surface; provides one reaction.
- **Fixed support:** Restrains translation and rotation; provides horizontal, vertical reactions and moment.



Common beam types: simply supported, cantilever, and overhanging with support symbols.

> **Malayalam support:** ബീമുകൾ അവയുടെ താങ്ങുകളുടെ (supports) അടിസ്ഥാനത്തിൽ simply supported, cantilever, overhanging എന്നിങ്ങനെ തരംതിരിക്കുന്നു. ലോഡുകൾ point load, UDL, UVL എന്നിങ്ങനെ വ്യത്യസ്ത തരത്തിലാകാം.

▼Determine reactions of simply supported beams under point load, UDL and combination

Simply supported beam: Reactions at supports are determined using the conditions of static equilibrium:

- $\Sigma F_x = 0$ (sum of horizontal forces = 0)
- $\Sigma F_y = 0$ (sum of vertical forces = 0)
- $\Sigma M = 0$ (sum of moments about any point = 0)

For a simply supported beam with point load P at distance a from left support (span L):

$R_A = P \cdot (L - a) / L$ (left reaction)

$R_B = P \cdot a / L$ (right reaction)

For UDL of w kN/m over full span L:

$R_A = R_B = wL / 2$

Worked example (point load):
Given: Simply supported beam span 6 m, point load 80 kN at 2 m from left.
Required: Support reactions.
Calculation:

$R_A = 80 \times (6-2)/6 = 80 \times 4/6 = 53.33$ kN

$R_B = 80 \times 2/6 = 26.67$ kN

Answer: $R_A = 53.33$ kN, $R_B = 26.67$ kN.

Worked example (UDL):
Given: Simply supported beam span 5 m, UDL of 12 kN/m over full span.
Required: Support reactions.
Calculation: Total load = 12 × 5 = 60 kN

$R_A = R_B = 60/2 = 30$ kN

Answer: $R_A = 30$ kN, $R_B = 30$ kN.

Worked example (combination):
Given: Span 4 m, point load 40 kN at 1 m from left, UDL 10 kN/m over entire span.
Required: Reactions.
Calculation:
Total UDL = 10 × 4 = 40 kN (acts at centre, 2 m from left)
ΣM about A = 0: $R_B \times 4 - 40\times1 - 40\times2 = 0 \Rightarrow 4R_B = 120 \Rightarrow R_B = 30$ kN
$\Sigma F_y = 0$: $R_A + 30 - 40 - 40 = 0 \Rightarrow R_A = 50$ kN
Answer: $R_A = 50$ kN, $R_B = 30$ kN.

Malayalam support: Simply supported beam-ൽ reactions കണ്ടുപിടിക്കാൻ static equilibrium conditions (ΣF=0, ΣM=0) ഉപയോഗിക്കുന്നു. Point load, UDL എന്നിവയ്ക്ക് പ്രത്യേകം ഫോർമുലകളുണ്ട്.

▾Determine reactions of cantilever beams under point load, UDL and combination

Cantilever beam: Fixed at one end and free at the other. The fixed support provides both a reaction force and a reaction moment.

For a cantilever with point load P at free end (length L):
Reaction at fixed end: $R = P$ (vertical)
Moment at fixed end: $M = P \times L$ (clockwise)

For UDL of w kN/m over full length L:
Reaction at fixed end: $R = wL$
Moment at fixed end: $M = wL^2 / 2$

Worked example (point load):
Given: Cantilever length 3 m, point load 60 kN at free end.
Required: Reaction and moment at fixed end.
Calculation: R = 60 kN, M = 60 × 3 = 180 kN·m
Answer: R = 60 kN (upward), M = 180 kN·m (clockwise).

Worked example (UDL):
Given: Cantilever length 4 m, UDL 15 kN/m.
Required: Reaction and moment.
Calculation: Total load = 15 × 4 = 60 kN, R = 60 kN
$M = 15 \times 4^2 / 2 = 15 \times 16 / 2 = 120$ kN·m
Answer: R = 60 kN, M = 120 kN·m.

Worked example (combination):
Given: Cantilever length 5 m, UDL 8 kN/m over full length, point load 30 kN at free end.
Required: Reaction and moment.
Calculation:
Total UDL = 8 × 5 = 40 kN → R_{UDL} = 40 kN, M_{UDL} = 8×25/2 = 100 kN·m
Point load: R_P = 30 kN, M_P = 30×5 = 150 kN·m
Total R = 40 + 30 = 70 kN; Total M = 100 + 150 = 250 kN·m
Answer: R = 70 kN, M = 250 kN·m.

Malayalam support: Cantilever beam-ൽ fixed end-ൽ reaction force ഉം reaction moment ഉം ഉണ്ടാകും. Point load, UDL എന്നിവയുടെ combination-ൽ ഓരോന്നിന്റെയും effect കൂട്ടിച്ചേർക്കണം.

▼Describe the principles of friction in various conditions

Friction: The resisting force that opposes relative motion between two surfaces in contact.

- **Static friction (F_s):** Acts before motion starts. Maximum value = $\mu_s \times N$.
- **Kinetic (dynamic) friction (F_k):** Acts during motion. Generally less than static friction. $F_k = \mu_k \times N$.
- **Limiting friction:** The maximum static friction just before motion begins.

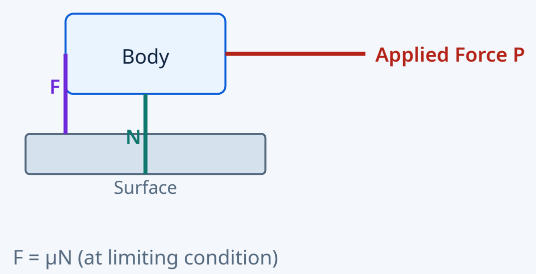
Friction laws:

$$F_{max} = \mu_s \cdot N \quad \text{(static)}$$

$$F_k = \mu_k \cdot N \quad \text{(kinetic)}$$

μ = coefficient of friction, N = normal reaction.

Angle of friction (φ): The angle between the normal reaction and the resultant of normal reaction and limiting friction. tan φ = μ.



Friction opposes the applied force; F = μN at the limiting condition.

Malayalam support: ഘർഷണം (friction) എന്നത് രണ്ട് പ്രതലങ്ങൾ തമ്മിലുള്ള ആപേക്ഷികചലനത്തെ എതിർക്കുന്ന ബലമാണ്. Static, kinetic എന്നിങ്ങനെ രണ്ടുതരം ഘർഷണമുണ്ട്. F = μN എന്നതാണ് അടിസ്ഥാന സമവാക്യം.

💡 **Module II key tip:** Always draw the FBD first. For beams, remember that the sum of moments about any point is zero — choose the point that eliminates the most unknowns. For friction, distinguish between static (before motion) and kinetic (during motion) cases.

MODULE III

12 hrs | CO3 | Apply

Distributed Areas & Masses

Centroid & CG | Composite figures | Moment of inertia | Polar moment of inertia

This module deals with the properties of distributed areas and masses — centroid, centre of gravity, and moment of inertia — which are essential for analysing the stability and strength of structural elements.

▼Recall formulae for finding centroid and centre of gravity of different geometrical plane figures and simple solids

Centroid: The geometric centre of a plane figure — the point where the total area is considered to be concentrated.
Centre of Gravity (CG): The point where the total weight of a body is considered to act. For uniform solids, CG coincides with the centroid.

Rectangle ($b \times h$):
Centroid: $(b/2, h/2)$
Area = $b \cdot h$

Triangle (base b, height h):
Centroid: $(b/3, h/3)$ from base
Area = $\frac{1}{2} b \cdot h$

Circle (radius r):
Centroid: at centre (0,0)
Area = πr^2

Semicircle (radius r):
Centroid: at $(0, 4r/3\pi)$ from base
Area = $\frac{1}{2}\pi r^2$

Sphere (radius r):
CG: at centre
Volume = $4/3\ \pi r^3$

Cylinder (radius r, height h):
CG: at $(r/2, h/2)$ from base
Volume = $\pi r^2 h$

Malayalam support: Centroid എന്നത് ഒരു ജ്യാമിതീയ രൂപത്തിന്റെ കേന്ദ്രബിന്ദുവാണ്. Centre of Gravity എന്നത് ഭാരം പ്രവർത്തിക്കുന്ന ബിന്ദുവാണ്. ഏകീകൃത വസ്തുക്കൾക്ക് ഇവ രണ്ടും ഒന്നുതന്നെയാകും.

▾Locate centroid of composite figures and centre of gravity of composite solids

Composite figures: Made up of two or more basic geometric shapes. The centroid is found by taking the weighted average of the centroids of each part.

For composite areas:

$$\bar{x} = (\Sigma A_i \cdot \bar{x}_i) / \Sigma A_i$$

$$\bar{y} = (\Sigma A_i \cdot \bar{y}_i) / \Sigma A_i$$

A_i = area of part i; $(\bar{x}_i, \bar{y}_i)$ = centroid of part i.

Common composite sections: L, T, C, I — these are built from rectangles. The procedure:

- Divide into simple shapes (rectangles, triangles, circles).
- Find the area and centroid of each part.
- Apply the composite formula.
- Choose a reference axis (usually the bottom or left edge).

Worked example (L-section):
Given: L-shaped section: vertical leg 100×20 mm, horizontal leg 80×20 mm.
Required: Centroid position $(\bar{x}, \bar{y})$ from bottom-left corner.
Calculation:
Part 1 (vertical): $A_1 = 100\times20 = 2000$ mm², $\bar{x}_1 = 10$, $\bar{y}_1 = 50$
Part 2 (horizontal): $A_2 = 80\times20 = 1600$ mm², $\bar{x}_2 = 40$, $\bar{y}_2 = 90$
$\Sigma A = 3600$ mm²
$\bar{x} = (2000\times10 + 1600\times40) / 3600 = (20000 + 64000) / 3600 = 23.33$ mm
$\bar{y} = (2000\times50 + 1600\times90) / 3600 = (100000 + 144000) / 3600 = 67.78$ mm
Answer: Centroid at (23.33 mm, 67.78 mm) from bottom-left.

Worked example (composite solid):
Given: A solid cylinder (radius 50 mm, height 100 mm) with a hemispherical cap (radius 50 mm) on top.
Required: CG from the base.
Calculation:
Cylinder: $V_1 = \pi(50)^2\times100 = 785{,}398$ mm³, $CG_1 = 50$ mm from base
Hemisphere: $V_2 = ⅔\pi(50)^3 = 261{,}799$ mm³, $CG_2 = 100 + (3\times50/8) = 118.75$ mm from base
Total V = 1,047,197 mm³
CG = (785398×50 + 261799×118.75) / 1047197 = (39269900 + 31088600) / 1047197 = 67.2 mm
Answer: CG at 67.2 mm from the base.

Malayalam support: സംയുക്ത രൂപങ്ങളുടെ centroid കണ്ടുപിടിക്കാൻ ഓരോ ഭാഗത്തിന്റെയും area, centroid എന്നിവ കണ്ടെത്തി weighted average എടുക്കുന്നു. L, T, C, I എന്നീ സെക്ഷനുകൾ പ്രധാനപ്പെട്ടവയാണ്.

▼Outline moment of inertia of plane lamina and solid bodies

Moment of Inertia (I): A measure of a body's resistance to angular acceleration. For a plane lamina, it is the second moment of area about an axis.

Moment of inertia about centroidal axes:

$I_x = \int y^2\, dA$ (about x-axis)

$I_y = \int x^2\, dA$ (about y-axis)

Parallel Axis Theorem: If the moment of inertia about the centroidal axis is known, the MOI about any parallel axis is:

$$I = I_c + A \cdot d^2$$

I_c = MOI about centroidal axis, A = area, d = distance between axes.

Rectangle (b×h):

$I_x = bh^3/12$ (about centroidal x-axis)

$I_y = hb^3/12$

Circle (radius r):

$I_x = I_y = \pi r^4/4$

Triangle (base b, height h):

$I_x = bh^3/36$ (about centroidal axis)

Solid cylinder (mass m, radius r):

$I = \frac{1}{2}mr^2$ (about longitudinal axis)

Malayalam support: Moment of inertia എന്നത് ഒരു വസ്തുവിന്റെ കോണീയ ത്വരണത്തിനെതിരായ പ്രതിരോധമാണ്. Parallel axis theorem ഉപയോഗിച്ച് centroidal axis-ൽ നിന്ന് മറ്റേതെങ്കിലും axis-ലേക്ക് MOI കണ്ടെത്താം.

▼Determine moment of inertia of composite figures and outline polar moment of inertia

Composite MOI: For a figure composed of multiple shapes, the total moment of inertia is the sum of the MOI of each part about the same axis (using the parallel axis theorem as needed).

For composite sections:

$$I = \Sigma (I_{ci} + A_i \cdot d_i^2)$$

I_{ci} = centroidal MOI of part i, d_i = distance from centroid of part i to the reference axis.

Worked example (I-section):
Given: I-section: flanges 120×20 mm, web 160×20 mm. Total height 200 mm.
Required: I_x about the centroidal horizontal axis.
Calculation:
Top flange: $A_1 = 120\times20 = 2400$ mm², $\bar{y}_1 = 190$ mm from bottom
Bottom flange: $A_2 = 120\times20 = 2400$ mm², $\bar{y}_2 = 10$ mm from bottom
Web: $A_3 = 20\times160 = 3200$ mm², $\bar{y}_3 = 100$ mm from bottom
Total A = 8000 mm²; $\bar{y} = (2400\times190 + 2400\times10 + 3200\times100) / 8000 = 100$ mm (symmetric)
$I_x = \Sigma(I_{ci} + A_i d_i^2)$
Top flange: $I_c = 120\times20^3/12 = 80{,}000$; d = 190−100 = 90; term = $80000 + 2400\times90^2 = 19{,}520{,}000$
Bottom flange: same = 19,520,000
Web: $I_c = 20\times160^3/12 = 6{,}826{,}667$; d = 0; term = 6,826,667
Total $I_x = 19{,}520{,}000 + 19{,}520{,}000 + 6{,}826{,}667 = 45{,}866{,}667$ mm⁴
Answer: $I_x = 45.87 \times 10^6$ mm⁴.

Polar Moment of Inertia (J): The moment of inertia about an axis perpendicular to the plane of the lamina. For a plane lamina:

$$J = I_x + I_y$$

For a circle: $J = \pi r^4/2$

Malayalam support: സംയുക്ത രൂപങ്ങളുടെ MOI കണ്ടെത്താൻ ഓരോ ഭാഗത്തിന്റെയും centroidal MOI, area, distance എന്നിവ ഉപയോഗിച്ച് parallel axis theorem പ്രയോഗിക്കുന്നു. Polar MOI എന്നത് ലംബമായ axis-നെ ആപേക്ഷിച്ചുള്ള MOI ആണ്.

💡 **Module III key tip:** Centroid and MOI calculations require careful division into simple shapes. Always draw the section, label dimensions, and choose a consistent reference axis. For composite sections, the parallel axis theorem is your main tool.

MODULE IV

10 hrs | CO4 | Understand

Design Thinking

Concept & stages | Empathize, Define, Ideate, Prototype, Test | Empathization techniques | Ideation tools & prototyping | Engineering applications

Design thinking is a human-centred approach to innovation that integrates the needs of people, the possibilities of technology, and the requirements for business success. This module introduces the five-stage process and its application in engineering.

▾Design Thinking: concept and stages

Design Thinking is a non-linear, iterative process that seeks to understand users, challenge assumptions, and redefine problems in an attempt to identify alternative strategies and solutions.
Five stages:

- **1. Empathize:** Understand the user's needs, feelings, and context. Conduct interviews, observations, and immersion.
- **2. Define:** Frame the problem clearly based on insights from the empathize stage. Create a problem statement.
- **3. Ideate:** Generate a wide range of creative ideas. Use brainstorming, mind mapping, SCAMPER, and other techniques.
- **4. Prototype:** Build tangible representations of ideas — from simple sketches to functional models. Prototypes are meant to be tested and improved.
- **5. Test:** Validate prototypes with real users. Gather feedback, learn what works, and iterate.

Iterative process — return to earlier stages as needed

The five stages of design thinking: Empathize → Define → Ideate → Prototype → Test.

> **Malayalam support:** ഡിസൈൻ തിങ്കിംഗ് എന്നത് ഉപഭോക്താവിനെ കേന്ദ്രീകരിച്ചുള്ള ഒരു നവീകരണ സമീപനമാണ്. അഞ്ച് ഘട്ടങ്ങൾ: Empathize, Define, Ideate, Prototype, Test. ഇത് ആവർത്തനപരമായ (iterative) ഒരു പ്രക്രിയയാണ്.

▼Empathization Techniques, Design Thinking as Divergent-Convergent Questioning

Empathization techniques: Methods to deeply understand the user's needs and context.

- **User interviews:** One-on-one conversations to uncover needs, pain points, and desires.
- **Observation:** Watching users in their natural environment to see how they interact with products or services.
- **Empathy mapping:** A visual tool that captures what a user says, thinks, does, and feels.
- **Journey mapping:** Tracking the user's experience across touchpoints to identify pain points and opportunities.
- **Persona creation:** Building fictional characters that represent user archetypes.

Divergent-Convergent Questioning:

- **Divergent thinking:** Generating many possible ideas, solutions, or perspectives. "How might we...?" questions open up possibilities.
- **Convergent thinking:** Narrowing down options, evaluating and selecting the best ones. "Which idea best meets the user's need?"
- Design thinking alternates between divergent (expanding) and convergent (focusing) modes at each stage.

> **Malayalam support:** Empathization techniques ഉപയോഗിച്ച് ഉപഭോക്താവിന്റെ ആവശ്യങ്ങൾ മനസ്സിലാക്കുന്നു. Divergent thinking ൽ നിരവധി ആശയങ്ങൾ സൃഷ്ടിക്കുകയും convergent thinking ൽ അവയിൽ നിന്ന് ഏറ്റവും മികച്ചത് തിരഞ്ഞെടുക്കുകയും ചെയ്യുന്നു.

▼Ideation Tools; Prototyping Methods

Ideation tools: Techniques to generate creative ideas.

- **Brainstorming:** Group idea generation without criticism — quantity over quality.
- **Mind mapping:** Visual diagram that organizes ideas around a central concept.
- **SCAMPER:** Substitute, Combine, Adapt, Modify, Put to another use, Eliminate, Reverse — a checklist for creative thinking.
- **Six Thinking Hats:** Looking at a problem from six different perspectives (logical, emotional, creative, etc.).
- **Sketching & storyboarding:** Visualizing ideas through quick drawings and sequences.

Prototyping methods: Building low-fidelity to high-fidelity representations of ideas.

- **Paper prototypes:** Hand-drawn sketches of interfaces or products — quick and cheap.
- **Digital mockups:** Using software (e.g., Figma, Adobe XD) to create interactive prototypes.
- **Physical models:** Using materials like cardboard, clay, foam, or 3D printing to create tangible objects.
- **Wizard of Oz:** A human simulates the system's behaviour behind the scenes.
- **Role-playing:** Acting out scenarios to experience the user's journey.

> **Malayalam support:** Ideation tools ആശയങ്ങൾ സൃഷ്ടിക്കാൻ സഹായിക്കുന്നു. Prototyping methods ആശയങ്ങളെ tangible രൂപത്തിൽ അവതരിപ്പിച്ച് പരീക്ഷിക്കാൻ അനുവദിക്കുന്നു.

▼Applications of design thinking in engineering domain

Design thinking has wide-ranging applications across the engineering domain. Some key areas include:

- **Product design & development:** Creating user-centred products that meet real needs — from consumer electronics to industrial machinery.
- **Process improvement:** Redesigning manufacturing, logistics, and service processes to enhance efficiency and user experience.
- **Systems engineering:** Designing complex systems (e.g., transportation, energy, communication) with a focus on user needs and system-level thinking.
- **Software & UX design:** Developing intuitive interfaces and digital experiences through iterative prototyping and user testing.
- **Sustainable engineering:** Designing for circular economy, energy efficiency, and environmental impact reduction.
- **Healthcare engineering:** Creating medical devices, diagnostic tools, and healthcare delivery systems that improve patient outcomes.
- **Civil & structural engineering:** Designing infrastructure (bridges, buildings, water systems) with consideration for community needs and resilience.

> **Engineering application example:**
> **Problem:** Farmers in a rural area need an affordable, low-maintenance water pump.
> **Empathize:** Interview farmers, observe their daily routines, understand their constraints (cost, electricity availability, maintenance skills).
> **Define:** "How might we design a solar-powered water pump that is affordable, easy to install, and requires minimal maintenance?"
> **Ideate:** Brainstorm various pump designs, energy sources, and materials. Consider off-the-shelf components.
> **Prototype:** Build a simple prototype using a small solar panel and a DC pump.
> **Test:** Install the prototype on a farm, gather feedback on performance, durability, and ease of use. Iterate based on learnings.

> **Malayalam support:** എൻജിനീയറിംഗ് മേഖലയിൽ design thinking പ്രയോഗിച്ച് ഉപഭോക്തൃകേന്ദ്രീകൃത ഉൽപ്പന്നങ്ങൾ, പ്രക്രിയകൾ, സംവിധാനങ്ങൾ എന്നിവ വികസിപ്പിക്കാം. ഇത് നവീകരണത്തിനും സുസ്ഥിരതയ്ക്കും വഴിയൊരുക്കുന്നു.

> 💡 **Module IV key tip:** Design thinking is not a linear checklist — it's an iterative mindset. Emphasize user empathy, embrace failure as learning, and always prototype to test assumptions. The five stages provide a structured framework, but the real value lies in the mindset.

📐 Formula Bank

Resolution:

$F_x = F \cos \theta$

$F_y = F \sin \theta$

Resultant:

$R = \sqrt{(\Sigma F_x^2 + \Sigma F_y^2)}$

$\alpha = \tan^{-1}(\Sigma F_y/\Sigma F_x)$

Moment:

$M = F \times d$

Lami's theorem:

$F_1/\sin\alpha = F_2/\sin\beta = F_3/\sin\gamma$

Simply supported (point load):

$R_A = P(L-a)/L$

$R_B = Pa/L$

Simply supported (UDL):

$R_A = R_B = wL/2$

Cantilever (point load at end):

$R = P$, $M = P \times L$

Cantilever (UDL):

$R = wL$, $M = wL^2/2$

Friction:

$F_{max} = \mu_s N$

$F_k = \mu_k N$

Composite centroid:

$\bar{x} = \Sigma A_i \bar{x}_i / \Sigma A_i$

$\bar{y} = \Sigma A_i \bar{y}_i / \Sigma A_i$

Parallel axis theorem:

$I = I_c + Ad^2$

Rectangle MOI:

$I_x = bh^3/12$

$I_y = hb^3/12$

Circle MOI:

$I_x = I_y = \pi r^4/4$

Polar MOI:

$J = I_x + I_y$

$J_{circle} = \pi r^4/2$

🖼️ Diagram Library for Rapid Revision

Force Systems
Concurrent, coplanar, parallel, collinear
↗ View diagram

Moment of Force
Turning effect: M = F × d
↗ View diagram

Lami's Theorem
Three forces in equilibrium
↗ View diagram

Beam Types & Supports
Simply supported, cantilever, overhanging
↗ View diagram

Friction Principles
F = μN at limiting condition
↗ View diagram

Design Thinking Stages
Empathize → Define → Ideate → Prototype → Test
↗ View diagram

🧪 Self-Learning Activities

The following activities are suggested for self-learning, with a total of **87.5 hours** across all modules.

Module I (22 hrs): Define different force systems; resolve forces and find resultant; compute moment of a force; solve simple problems from free body concept and Lami's theorem.

Module II (22 hrs): Identify different types of beams, loads and supports; determine reactions of simply supported and cantilever beams under point load, UDL and combination; describe principles of friction.

Module III (22 hrs): Recall centroid/CG formulae; locate centroid of composite figures (L, T, C, I) and CG of composite solids; outline moment of inertia; determine MOI of composite figures; outline polar moment of inertia.

Module IV (21.5 hrs): Design Thinking concept and stages; empathization techniques; divergent-convergent questioning; ideation tools; prototyping methods; applications in engineering.

📝 Assessment Methodology

CIE & ESE Pattern

Assessment breakdown

Mode	Details	Marks
Attendance	—	5
CA1	Self-learning activities (Module 1), 2 hrs	15
CA2	Self-learning activities (Module 2), 2 hrs	15
CA3	Self-learning activities (Module 3), 2 hrs	15
CA4	Self-learning activities (Module 4), 2 hrs	15
CA5	Series Exam (2 modules), 2.5 hrs	50
CA6	Series Exam (2 modules), 2.5 hrs	50
ESE	Written Exam (Theory), 2.5 hrs	60

CIE total = 40 (converted from CA marks), ESE = 60. Tentative schedule: CA1–CA4 by end of semester; CA5 at 8th week; CA6 at 15th week; ESE at end of semester.

End Semester Examination Pattern

Duration: 2.5 hours | **Total Marks:** 60

ESE question paper pattern

Part	Type of Questions	Marks per question	Number of questions	Total Marks
Part A	2 questions from each module (1 mark each)	1	8	8
Part B	2 out of 3 questions from each module (3 marks each)	3	8	24
Part C	1 set question with choice from each module (7 marks each)	7	4	28
			Total	**60**

❓ Module-wise Questions with Answers

Module I

Q: What are the different types of force systems?

A: Force systems are classified as: concurrent (lines of action meet at a point), coplanar (all forces lie in the same plane), collinear (forces act along the same line), parallel (lines of action are parallel), and non-concurrent/non-parallel (general case).

Q: State Lami's theorem.

A: If three forces acting at a point are in equilibrium, then each force is proportional to the sine of the angle between the other two forces: $F_1/\sin\alpha = F_2/\sin\beta = F_3/\sin\gamma$.

Q: Explain the free body diagram (FBD) concept.

A: A free body diagram is a sketch of a body isolated from its surroundings, showing all external forces acting on it. It helps in applying equilibrium equations ($\Sigma F=0$, $\Sigma M=0$) to solve mechanics problems.

Module II

Q: Distinguish between simply supported and cantilever beams.

A: Simply supported beam is supported at both ends (pin and roller), allowing rotation but not translation. Cantilever beam is fixed at one end and free at the other, providing both reaction force and moment at the fixed end.

Q: Derive the reactions for a simply supported beam with a point load P at distance a from left support.

A: Taking moments about left support: $R_B \times L = P \times a \rightarrow R_B = Pa/L$. From $\Sigma F_y=0$: $R_A = P - R_B = P(L-a)/L$.

Q: Define static and kinetic friction.

A: Static friction is the resisting force that prevents motion before it starts; its maximum value is $F_{max} = \mu_s N$. Kinetic (dynamic) friction acts during motion and is $F_k = \mu_k N$, generally less than static friction.

Module III

Q: What is the centroid of a composite figure? How is it found?

A: The centroid of a composite figure is the weighted average of the centroids of its component parts: $\bar{x} = \Sigma A_i\bar{x}_i/\Sigma A_i$, $\bar{y} = \Sigma A_i\bar{y}_i/\Sigma A_i$. Each part's area and centroid must be known.

Q: State the parallel axis theorem.

A: The moment of inertia of a body about any axis is equal to the sum of its moment of inertia about a parallel centroidal axis and the product of its area and the square of the distance between the axes: $I = I_C + Ad^2$.

Q: Write the formula for polar moment of inertia of a circle.

A: $J = \pi r^4/2$, where r is the radius of the circle. It is the sum of I_x and I_y (which are equal for a circle).

Module IV

Q: List the five stages of design thinking.

A: The five stages are: 1. Empathize, 2. Define, 3. Ideate, 4. Prototype, 5. Test. The process is iterative and non-linear.

Q: What is the difference between divergent and convergent thinking?

A: Divergent thinking involves generating many creative ideas and possibilities (expanding). Convergent thinking involves narrowing down options, evaluating, and selecting the best solution (focusing). Design thinking alternates between both.

Q: Give two applications of design thinking in engineering.

A: 1. Product design: creating user-centred consumer electronics, medical devices, or industrial machinery. 2. Process improvement: redesigning manufacturing or service workflows to enhance efficiency and user experience.

📄 Practice Model Question Paper

Based on the official ESE pattern — not an official SITTTR model question paper.

Duration: 2.5 hours | **Total Marks:** 60

Part A (8 × 1 = 8 marks)

1. Define a coplanar force system.
2. Write the formula for the moment of a force.
3. What is a simply supported beam?
4. State the relationship between friction and normal reaction.

5. Define centroid of a plane figure.
6. Write the formula for the moment of inertia of a rectangle about its centroidal axis.
7. What is the first stage of design thinking?
8. Give one ideation tool.

Part B (8 × 3 = 24 marks)

Answer any two out of three from each module.
Module I: 9. Explain the resolution of a force into horizontal and vertical components. 10. State Lami's theorem with a diagram. 11. A force of 120 N acts at 45° to the horizontal. Find its components.
Module II: 12. Differentiate between pin, roller, and fixed supports. 13. A simply supported beam of span 4 m carries a UDL of 10 kN/m. Find reactions. 14. Explain the angle of friction.
Module III: 15. Find the centroid of a T-section (flange 100×20, web 80×20). 16. State the parallel axis theorem. 17. Calculate I_x for a circle of radius 50 mm.
Module IV: 18. Explain the empathize stage of design thinking. 19. Describe the SCAMPER ideation technique. 20. Give two engineering applications of design thinking.

Part C (4 × 7 = 28 marks)

Answer one set question with choice from each module.
Module I: 21. a) Define a force system and classify it. b) Two forces of 80 N and 60 N act at 60° to each other. Find the resultant. *OR* 22. a) Explain the free body diagram concept. b) Using Lami's theorem, find the unknown force when F_1=100 N, F_2=80 N, and angles α=120°, β=100°.
Module II: 23. a) Derive the reaction formulae for a simply supported beam with a point load. b) A cantilever beam of 3 m carries a point load of 50 kN at the free end. Find reaction and moment. *OR* 24. a) Describe the principles of static and kinetic friction. b) A block of weight 200 N rests on a surface with μ_s=0.4. Find the limiting friction.
Module III: 25. a) Find the centroid of an L-section (vertical 120×20, horizontal 80×20). b) Calculate the moment of inertia of the same L-section about its centroidal x-axis. *OR* 26. a) Explain the parallel axis theorem with a diagram. b) Find the polar moment of inertia of a solid cylinder of radius 60 mm.
Module IV: 27. a) Explain the five stages of design thinking. b) Describe empathization techniques with examples. *OR* 28. a) Discuss divergent-convergent questioning in design thinking. b) Give a detailed application of design thinking in engineering product development.

⚡ Quick Revision

Module I

- Force system types
- Resolution: F_x=Fcosθ, F_y=Fsinθ
- Resultant: R = $\sqrt{(\Sigma F_x^2+\Sigma F_y^2)}$
- Moment: M = F×d
- Lami: $F_1/\sin\alpha = F_2/\sin\beta = F_3/\sin\gamma$

Module II

- Simply supported: R_A=P(L−a)/L

- UDL: $R_A=R_B=wL/2$
- Cantilever: $R=P$, $M=P\times L$
- Cantilever UDL: $R=wL$, $M=wL^2/2$
- Friction: $F=\mu N$

Module III

- Centroid: $\bar{x}=\Sigma A_i\bar{x}_i/\Sigma A_i$
- Parallel axis: $I = I_c + Ad^2$
- Rectangle $I_x = bh^3/12$
- Circle $I_x = \pi r^4/4$
- Polar $J = I_x + I_y$

Module IV

- 5 stages: Empathize, Define, Ideate, Prototype, Test
- Empathize: interviews, observation, empathy mapping
- Ideate: brainstorming, SCAMPER, mind maps
- Prototype: paper, digital, physical models
- Applications: product design, process improvement

Common Mistakes to Avoid

FBD errors: Forgetting to include all forces or mislabelling directions.
Sign convention: Inconsistent use of positive/negative signs for moments.
UDL conversion: Forgetting that UDL acts at the centre of the distributed length.
Composite centroid: Using wrong reference axis or forgetting to subtract holes.
Parallel axis theorem: Mixing up centroidal axis and reference axis distances.
Design thinking: Treating it as a linear process — remember it's iterative.

Exam Checklist

✅ Review all formula cards
✅ Practice at least one worked example per topic
✅ Draw FBDs for every problem
✅ Check units and sign conventions
✅ For design thinking, remember the five stages and key techniques
✅ Go through module-wise Q&A
✅ Attempt the model paper under timed conditions

📖 Glossary

Important terms and their meanings

Term	Meaning
Force	A push or pull that changes or tends to change the state of motion of a body.
Resultant	A single force that produces the same effect as a system of forces.
Moment	The turning effect of a force about a point; $M = F \times d$.
Free Body Diagram	A diagram of a body isolated from its surroundings, showing all external forces.
Lami's theorem	Relates three forces in equilibrium: $F_1/\sin\alpha = F_2/\sin\beta = F_3/\sin\gamma$.
Beam	A structural element that carries loads by bending.
UDL	Uniformly Distributed Load — load spread evenly over a length.
Friction	Resisting force that opposes relative motion between surfaces.
Centroid	The geometric centre of a plane figure.
Centre of Gravity	The point where the total weight of a body is considered to act.
Moment of Inertia	Measure of a body's resistance to angular acceleration.
Parallel Axis Theorem	$I = I_c + Ad^2$ — relates MOI about any axis to the centroidal axis.
Polar Moment of Inertia	MOI about an axis perpendicular to the plane; $J = I_x + I_y$.
Design Thinking	Human-centred, iterative approach to innovation and problem-solving.
Empathize	Understand the user's needs and context.
Ideate	Generate creative ideas using brainstorming, SCAMPER, etc.
Prototype	A tangible representation of an idea for testing and iteration.

📚 References

Textbooks

Prescribed textbooks

Sl. No.	Author	Title	Publication
1	Ramamrutham S.	Engineering Mechanics	S. Chand & Co., New Delhi
2	Bansal R K	A text book of Engineering Mechanics	Laxmi Publications
3	Khurmi R.S.	Applied Mechanics	S. Chand & Co., New Delhi
4	E Balaguruswamy, Bindhu Vijayakumar	Design thinking: A beginners perspective (ISBN-13: 978-9355329578)	McGraw Hill Publications
5	Shalini Rahul Tiwari, Rohit Rajendra Swarup	Design Thinking: A Comprehensive Textbook (ISBN-13: 978-9357468640)	Wiley India Private Limited

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Reference books

Sl. No.	Author	Title	Publication
1	Ram, H. D.; Chauhan, A. K	Foundations and Applications of Applied Mechanics	Cambridge University Press
2	Meriam, J. L., Kraige, L.G	Engineering Mechanics-Statics, Vol. I	Wiley Publication, New Delhi
3	Pahl, G., Beitz, W., Feldhusen, J., Grote, K.-H	Engineering Design: A Systematic Approach, Third Edition (ISBN 978-1-84628-319-2)	Springer 2007
4	Chandramouli Subramanian, Sankaran Venkataramani	Design Thinking: A Hands-on Approach (ISBN-13: 978-8197424922)	Orient Blackswan Pvt Ltd
5	Pavan Soni	Design your thinking (ISBN-13: 978-0670094097)	Penguin Random House India PVT Ltd.

Web Resources

NPTEL — Engineering Mechanics (Modules I & II)
NPTEL — Mechanics (Modules I, II & III)
NPTEL — Design Thinking (Module IV)